Prajwal Kumar K

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LinkedIn | Portfolio | GitHub | Codeforces | CodeChef | LeetCode | AtCoder

EDUCATION

Indian Institute of Information Technology, Kottayam — Kottayam, Kerala
Bachelor of Technology in Computer Science and Engineering; CGPA: 8.66/10 — *Aug 2024 – May 2028 (expected)*

TECHNICAL SKILLS

Programming: C++, Rust, Python, TypeScript, JavaScript, Java, SQL, Solidity, MATLAB
AI/ML: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib
Research and engineering: Jupyter, Simulink, FastAPI, Next.js
Research interests: Algorithms and systems, game theory, machine learning and NLP, AI safety
Languages: English (Fluent), German, Hindi, Kannada

EXPERIENCE

Coders Club Coordinator — Jan 2025 – Present
IIIT Kottayam — *Kottayam, Kerala*

- Organized contests for 80+ participants, curated problems, and configured Codeforces Gym rounds.

Openverse Technical Lead (Open Source Club) — May 2025 – Jul 2026
IIIT Kottayam — *Kottayam, Kerala*

- Led technical work and pull request reviews for 50+ Openverse contributors.
- Co-built Algorithima, a visual teaching tool used by professors for DSA and C programming courses.

Full-Stack Web Developer Intern — Dec 2025 – Mar 2026
CIO Mogul (Magazine) — *Remote*

- Built 3 magazine websites from scratch with a teammate: CIO Mogul, CIO Mogul India, and CIO Healthcare.
- Developed publishing dashboards with Next.js, React, and PostgreSQL on Supabase; automated PDF cover extraction and split double-page spreads for the magazine reader.

PROJECTS

PokeForge | *Python, PyTorch, Gen 9 OU* — Code

- Built MLP and Transformer policies with legal-action masks and 61 offline contract tests; prototyped evaluation against pinned FoulPlay and Metamon baselines (simulator conformance in progress).

Bluff | *Python, PyTorch, Bayesian modeling* — Live | Code

- Built an imperfect-information card-game testbed with Bayesian opponent models; a bluff detector improved draw-aware score by 4.9–5.8 percentage points across 8 synthetic profiles versus a no-detector baseline.

SiliconSpire | *C++, Rust, metaheuristics* — C++ | Rust

- Implemented Grey Wolf Optimization and Tabu Search for facility placement using the Quadratic Assignment Problem formulation.
- Matched published optima in 77/90 Rust runs versus 34/90 C++ runs across 3 small QAPLIB instances; heuristic results are instance- and seed-dependent.

KPlaceNet | *PyTorch, image geolocation, uncertainty* — Evaluation branch

- Contributed reproducible OSV-5M scaling and calibration evaluation to a team ResNet-50 geolocation project over 300 geographic cells.
- Across 3 seeds on one 3,000-image test sample, 20k vs. 10k training lowered mean error from 5,977 to 5,522 km and raised within-200-km accuracy from 3.76% to 5.30%; temperature scaling lowered ECE from 0.417 to 0.045 without changing accuracy.

BattleCP | *Rust, Axum, Tokio, WebSockets* — Live | Code

- Built and launched a multiplayer Battleship game where solving Codeforces problems earns turns; its Rust WebSocket server enforces game rules and submission checks. Recorded 6,922 cumulative page views.

ACHIEVEMENTS AND CERTIFICATIONS

Publication: First-listed author and presenter, "Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications," IEEE ICICI 2025. Paper.
Algorithima: selected for and presented at OOSC 3.0, IIT Kanpur (2025); team presentation at T4E 2025, IIT Madras.
Hackathon: Won Best Software Award at IEEE MACE .hack(); 2026 with Team Hogorithm (Anchor).
Programming: UNSW Battlecode 2026, equal 254th of 972 teams (Oct 2026; team bbn); IICPC 2026 team rank 3,352; ICPC 2025 team rank 2,273 (bnb); Codeforces Pupil (peak 1221); CodeChef peak 1463; 400+ problems solved.
Cyfrin Updraft (2025): Blockchain Basics, Rust Programming Basics, Solidity Smart Contract Development, and Fundamentals of Zero-Knowledge Proofs.